Meat intake and reproductive parameters among young men

Background In the United States, anabolic sex steroids are administered to cattle for growth promotion. There is concern regarding the reproductive consequences of this practice for men who eat beef. We investigated whether meat consumption was associated with semen quality parameters and reproductive hormone levels in young men. Methods Semen samples were obtained from 189 men aged 18-22 years. Diet was assessed with a previously validated food frequency questionnaire. We used linear regression to analyze the cross-sectional associations of meat intake with semen quality parameters and reproductive hormones, while adjusting for potential confounders. Results There was an inverse relation between processed red meat intake and total sperm count. The adjusted relative differences in total sperm counts for men in increasing quartiles of processed meat intake were 0 (ref), −3 (95% confidence interval = −67 to 37), −14 (−82 to 28), and −78 (−202 to −5) million (test for trend, P = 0.01). This association was strongest among men with abstinence time less than 2 days and was driven by a strong inverse relation between processed red meat intake and ejaculate volume (test for trend, P =0.003). Conclusions In our population of young men, processed meat intake was associated with lower total sperm count. We cannot distinguish whether this association is due to residual confounding by abstinence time or represents a true biological effect.